

Title: Hotel Rate Comparator using Temporal Workflows

Type: Full Stack (Node.js + TypeScript + Temporal SDK + React/Next.js)

Goal: Build a full-stack app where users can search hotels, with the backend powered by Temporal workflows to fetch and compare hotel rates from two suppliers.

Background

You are building a hotel search engine that finds the best rate by calling two external supplier APIs. These suppliers can have delays or errors. Use **Temporal.io** to orchestrate the process reliably — fetching, comparing, and returning the best rate.

You will also build a simple **React** frontend for submitting the search and showing results.

Requirements

1. Frontend (React or Next.js)

- A page with a form:
 - City
 - Check-in Date
 - Check-out Date
- Submit triggers a backend call.
- Show a loading state.
- On success: display hotel name, price, supplier.
- On failure: show a clear error message.

Use TypeScript. Minimal styling is fine

2. Backend (Node.js + Temporal)

- Use Temporal SDK (TypeScript).
- Backend endpoint: `/api/search-hotels`

- Workflow logic:
 - Call Supplier A and Supplier B (in parallel).
 - Handle errors/timeouts from suppliers.
 - Select the cheapest result.
 - Return it to the client.
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3. Mock Supplier APIs

- Build two endpoints: `/supplierA/hotels`, `/supplierB/hotels`
 - Return hotel list with `hotelId`, `name`, `price`
 - Simulate varying behaviors:
 - Delays
 - Timeouts
 - Empty responses
 - Server errors
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Scenario Tests

Write **unit tests** and **workflow tests** for the following scenarios:

Basic Scenarios

Scenario	Expected Outcome
Supplier A cheaper	Return A's result
Supplier B cheaper	Return B's result
Both return same rate	Pick deterministically (e.g., Supplier A)
Supplier A fails, B succeeds	Return B's result

Both fail	Return error
One returns empty	Use available result
Both return empty	Return "No hotels found"

Advanced Scenarios

Scenario	Expected Behavior
One supplier takes >5s to respond	Cancel slow activity, proceed with one result
Supplier A fails 2x before success	Still succeeds if within retry policy
User cancels the request mid-way	Workflow should stop gracefully

Submission Instructions

- GitHub repo with read access granted to kaushal@tripare.com
- Must include:
 - `README.md` with setup instructions
 - Scripts for running frontend, backend, and Temporal Worker
 - Tests with clear coverage of scenarios
 - Any known limitations or assumptions

Stack Constraints

Layer	Tech Stack
Frontend	React (TypeScript)
Backend	Node.js (TypeScript), Temporal SDK

Testing	Jest / Playwright / Your choice
Server APIs	Express.js or any framework you prefer